

Lesson 20 — Design Challenge — Tell a Story in Order

Matches LEGO Teacher Portal lesson: "Story Steps".

Objective (I can...): I can program a sequence that tells a little story, in the right order. **CS concept:** Programming a sequence to tell a story (applying ordered steps for a purpose) · **Builds on:** Lessons 16–19. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A storytelling app plays its scenes in order, one after another, to make sense. *Ask the group:* "What happens to a story if the ending plays first?"

Materials: LEGO CS & AI kit + device per pair, picture storyboards, story props. **Model already built** (Sequences model from Lesson 16) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Sequences review — pairs put a 4-scene story (beginning, middle, surprise, end) in order and tell it fast. (*Theme: Story theater.*) - **Concept + demo (8):** Challenge: code a sequence that acts out a tiny story with the machine (lights, motor moves, sounds) in order. Coach storyboards one idea, then hands off. (*Try "show a friend."*) - **Build/Code (25):** Pairs storyboard then code a 4+ step story sequence on the built model, running to make sure the scenes play in order. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add one surprise scene in the right spot and update the storyboard. - **Share + tidy-up (10): Story theater** — pairs perform their machine's story in order while the group watches. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator narrates the story beats and points to the storyboard; swap at the wiggle break. (*Trio rotates plan/code/perform.*)

Differentiation: *Support* — a 3-scene story with a partial storyboard. *Extension / fast-finisher* — a 6-scene story with a light, a sound, and a motor move. *Reteach* — reuse their Lesson 19 picture plan as a starting story.

Exit check: Pair's machine tells a 4+ scene story with the scenes in the right order, matching a storyboard. **Progress layer:** Send home the **Unit 4 parent note** ("your child can build, fix, and tell a story with an ordered program") with the **build photo**. **At-home (optional):** Storyboard a 4-scene story and act it out for your family in order. ~5 min.